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APPARATUSES AND METHODS FOR FACILITATING DETECTION OF COMMUNICATION SERVICE DEGRADATION IN COMMUNICATION NETWORKS AND SYSTEMS

Abstract

Aspects of the subject disclosure may include, for example, analyzing, by a processing system including a processor, at least first data from a voice communication session, generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold, identifying, by the processing system and based on the generating, a source of the degradation, and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource. Other embodiments are disclosed.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The subject disclosure relates to apparatuses and methods for facilitating detection of communication service degradation in communication networks and systems.

BACKGROUND

[0002] As the world increasingly becomes connected via vast communication networks and systems and via various communication devices, additional opportunities are created/generated to provision communication services. A voice call is an example of a communication service.

[0003] Call degradation is a pervasive problem in communication networks/systems, often resulting from factors such as network/system congestion, signal interference, noise, hardware failures, and suboptimal configurations. These issues can lead to reduced call quality, characterized by symptoms like dropped calls, audio distortion, and increased latency. Left unaddressed, call degradation can negatively impact the quality of the user experience and may result in customer/subscriber churn.

[0004] There has been a steady rise in use/adoption of encryption to protect customer/subscriber data, due in part to a changing regulatory landscape. Additionally, increased cloud computing adoption drives demand for security safeguards. For network/system operators and service providers, hidden/secured data reduces visibility/oversight, which makes it difficult to perform analytics for use cases such as fraud identification, network/system management and optimizations with capital forecasting, informative customer/subscriber usage reporting, as well as revenue-generating use cases explicitly permitted via opt-in such as relevant/targeted advertising. Sparse data leads to incomplete and/or inaccurate analyses, which in turn leads to flawed forecasting and usage reporting used for network/system and product/service planning and diminished monetization opportunities and revenues. In brief, there is an increasing need for network/system operators and service providers to advance analytics-based methods to regain visibility into encrypted communications and to ensure continuity of accurate and timely status capture and reporting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0006] FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

[0007] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system in accordance with various aspects described herein.

[0008] FIG. 2B illustrates a diagram of packet delay in accordance with aspects of this disclosure.

[0009] FIG. 2C illustrates a diagram of packet loss in accordance with aspects of this disclosure.

[0010] FIG. 2D depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0011] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0012] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0013] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0014] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0015] The subject disclosure describes, among other things, illustrative embodiments for detecting degradation in qualities of communication services and implementing modifications to remediate the degradation. Other embodiments are described in the subject disclosure.

[0016] One or more aspects of the subject disclosure include, in whole or in part, obtaining a model in respect of voice communications; obtaining data representative of a voice communication session; comparing the data to the model to generate a prediction that the data is subject to degradation; based on the prediction, identifying a remediation to address the degradation; and implementing the remediation.

[0017] One or more aspects of the subject disclosure include, in whole or in part, analyzing first data from a voice communication session; determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay; and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold.

[0018] One or more aspects of the subject disclosure include, in whole or in part, analyzing, by a processing system including a processor, at least first data from a voice communication session; generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold; identifying, by the processing system and based on the generating, a source of the degradation; and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.

[0019] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, the system **100** can facilitate, in whole or in part, obtaining a model in respect of voice communications, obtaining data representative of a voice communication session, comparing the data to the model to generate a prediction that the data is subject to degradation, based on the prediction, identifying a remediation to address the degradation, and implementing the remediation. The system **100** can facilitate, in whole or in part, analyzing first data from a voice communication session, determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay, and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold. The system **100** can facilitate, in whole or in part, analyzing, by a processing system including a processor, at least first data from a voice communication session, generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold, identifying, by the processing system and based on the generating, a source of the degradation, and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.

[0020] In particular, in FIG. 1 a communications network **125** is presented for providing broadband access **110** to a plurality of data terminals **114** via access terminal **112**, wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a

plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

[0021] The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VoIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0022] In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0023] In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0024] In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

[0025] In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

[0026] In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0027] In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0028] By way of introduction, aspects of this disclosure may facilitate techniques for enhancing qualities or characteristics of communication services provided by one or more network or system operators or service providers. In some embodiments, an analysis may be undertaken in respect of data, packets, signals, metadata, etc., to obtain knowledge of a status or health of a communication network or system in relation to a provisioning of one or more communication services. Such status or health may be generated in real-time, or in near real-time. In some embodiments, aspects of this

disclosure may be implemented as part of a background task or methodology to assess network/system health or status over time.

[0029] In respect of a voice call communication service, degradation in quality may be a result of packet loss and/or packet delay. Packet loss occurs when data packets sourced from a first communication device (or, analogously, a sender or speaker) are lost during transmission, primarily due to network/system congestion, signal interference, or hardware issues. When a significant number of packets are lost (e.g., in an amount greater than a threshold), a second communication device associated with a recipient user might not have enough data to reconstruct the associated audio stream, leading to gaps or missing sounds in conversation. This can render the speaker's voice/speech unintelligible, making it difficult for the listener/recipient to understand the speaker. Packet delay refers to the time it takes for voice data packets to travel from the sender (upon transmission) to the recipient. High latency (e.g., latency exceeding a threshold) can be caused by factors such as long transmission distances, network/system congestion, or inefficient routing. When packet delays are excessive (e.g., greater than a threshold), speech or conversation can become disjointed, with noticeable delays between when a speaker talks and when the listener hears the speaker. This can lead to a “robotic” sound, as the audio stream is played back at the receiving end with uneven timing.

[0030] Packet loss and packet delay may contribute to degraded call quality, but their impacts on the audio playback/presentation/rendering may be different. For example, packet loss typically results in missing or garbled sounds, making the conversation unintelligible; packet delay can cause a disjointed, robotic sound due to the uneven playback of the associated audio stream.

[0031] With the foregoing as an introduction to aspects of this disclosure, reference may now be made to FIG. 2A, which is a block diagram illustrating an example, non-limiting embodiment of a system **200a**. In some embodiments, one or more parts/portions of the system **200a** may function within, or may be operatively overlaid upon, one or more parts/portions of the system **100** of FIG. 1.

[0032] The system **200a** may include one or more networks, such as a core network **202a** and access networks (represented by a first tower or base station **206a-1** and a second tower or base station **206a-2**). The core network and the access network(s) may facilitate communication services in respect of one or more communication devices, such as for example a first user equipment (UE) **210a-1** and a second UE **210a-2**. Such communication services may, illustratively, include a voice call or voice communication service. Communications or signaling facilitated via the system **200a** may include a use of one or more communication channels, links, or the like. Such communications/signaling may occur in a unidirectional manner or fashion or may be bidirectional in nature. Aspects of such communications or signaling involving the various devices or entities shown in FIG. 2A are described in further detail below.

[0033] Referring now to FIG. 2B, a diagram **200b** demonstrating aspects of packet delay is shown. In particular, the diagram **200b** shows signal energy on the vertical axis relative to time on the horizontal axis. The diagram **200b** is effectively two diagrams in one, where a first portion of the diagram **200b** demonstrates the signal energy for a sender/sending device (e.g., UE **210a-1** of FIG. 2A) as a function of time and a second portion of the diagram **200b** demonstrates the signal energy for a recipient/receiving device (e.g., UE **210a-2** of FIG. 2A) as a function of time.

[0034] In the illustrative scenario depicted in FIG. 2B, the sender causes two data packets to be emitted, denoted by reference characters **204b-1** and **208b-1**. Due to factors contributing to packet delay (as described above, for example), the packets **204b-1** and **208b-1** may be received at/by the recipient some time later as demonstrated via reference characters **204b-2** and **208b-2**, respectively. The delay between the sending of the first packet **204b-1** by the sender and the receipt of the first packet **204b-2** by/at the recipient is represented by reference character **204b-3**. Similarly, the delay between the sending of the second packet **208b-1** by the sender and the receipt of the second packet **208b-2** by/at the recipient is represented by reference character **208b-3**.

[0035] In practice, a duration of the time delay **204b-3** and a duration of the time delay **208b-3** might not be the same or equal. For example, the duration of the time delay **204b-3** may be less than the duration of the time delay **208b-3**. In some embodiments, the duration of the time delay **204b-3** may be tolerable, which is to say that the time delay **204b-3** might not appreciably impact the quality of an associated voice call from the perspective of a user (e.g., the recipient). In contrast, the duration of time delay **208b-3** might not be tolerable, which is to say that the time delay **208b-3** may have a significant, negative impact on the quality of the voice call from the perspective of the user.

[0036] Referring now to FIG. 2C, a diagram **200c** demonstrating aspects of packet loss is shown. In particular, the diagram **200c** shows signal energy on the vertical axis relative to time on the horizontal axis. The diagram **200c** is effectively two diagrams in one, where a first portion of the diagram **200c** demonstrates the signal energy for a sender/sending device (e.g., UE **210a-1** of FIG. 2A) as a function of time and a second portion of the diagram **200c** demonstrates the signal energy for a recipient/receiving device (e.g., UE **210a-2** of FIG. 2A) as a function of time.

[0037] In the illustrative scenario depicted in FIG. 2C, the sender causes three data packets to be emitted, denoted by reference characters **204c-1**, **208c-1**, and **212c-1**. Due to factors contributing to packet loss (as described above, for example) one or more of the packets emitted by the sender might not be received by/at the recipient. For example, the first packet **204c-1** and the third packet **212c-1** may be received by/at the recipient (as demonstrated via reference characters **204c-2** and **212c-2**, respectively). Conversely, the second packet **208c-1** emitted by the sender might not be successfully received by/at the recipient (as denoted by the absence of the second packet at the recipient in FIG. 2C). The absence of the second packet at the recipient in FIG. 2C may (or might not) have a significant, negative impact on the quality of the voice call.

[0038] While illustrated separately in FIG. 2B and FIG. 2C for the sake of ease and convenience in description, aspects of both packet delay and packet loss may be present in a given embodiment. Stated differently, and based on the circumstances or conditions at hand, any given packet may potentially be the subject of packet delay and/or packet loss. Further, it is noted that packet loss in this context refers to an inability for a receiver/recipient to successfully receive and decode data associated with a packet. In this regard, even if a packet is received by/at a receiver, the packet might be deemed as having been lost if the characteristics of the associated signal are of such poor quality that the data associated with the packet cannot be recovered. Indeed, and as one of skill in the art will appreciate, propagation characteristics associated with a communication channel, link, or the like may cause a transformation (e.g., a distortion) to be imparted on a signal as the signal is conveyed from a source to a destination.

[0039] In some embodiments, modeling techniques may be used to model voice/speech communications. For example, experience and knowledge may indicate that users tend to speak with a certain rhythm or cadence, often speaking for a first portion or period of time, then pausing for a second portion or period of time, before resuming speech in a third portion or period of time. To demonstrate, a speaker will often pause between the conclusion of a first sentence and the start or initiation of a second sentence. Such patterns of behavior in speech may be exploited or leveraged to identify when packet degradation has likely been experienced by comparing packets associated with actual speech/voice to packets associated with expected speech/voice (relative to one or more tolerance bands or thresholds selected or established to allow for reasonable variations—e.g., a fast talker versus a slow talker). Still further, in some instances a communication network or system may be arranged or configured to transmit a certain number or count of packets over a given unit of time. For example, and assuming a network or system with 250 millisecond resolution, segmenting a voice call into data units (each unit hereinafter referred to as a CAU) of approximately 200 milliseconds may lend itself to an examination of the CAUs to potentially identify packet corruption or distortion (in an uplink direction and/or a downlink direction). For example, it may be assumed that each CAU has: (1) ten speech packets, (2) one idle packet, or (3)

two speech packets and one idle packet. An analysis may be performed relative to the CAUs to determine/identify patterns that indicate likely packet loss or delay, based on a deviation from the assumed norms.

[0040] Upon detecting likely packet loss or delay, a root cause analysis may be undertaken or performed in an effort to identify or determine the cause or source of the packet loss or delay. For example, and with reference to FIG. 2A, if a packet loss or delay were to occur between the first UE **210a-1** and the tower/base station **206a-1**, that may be indicative of an issue with signaling involving the tower/base station **206a-1** (similar remarks apply in respect of the second UE **210a-2** and the tower/base station **206a-2**). If the packet loss or delay were to occur between the tower/base station **206a-1** and the core network **202a**, that may be indicative of a server fault or error.

[0041] Once a root cause is identified, a recommendation or suggestion may be generated to address the root cause. For example, such a recommendation or suggestion may include, without limitation: adjusting a network/system configuration, reallocating resources, upgrading equipment, handing over a communication session involving a UE from a first resource to a second resource, adjusting a communication parameter involving a network/system resource or a UE (such as, for example, a transmission power level, a receiver sensitivity level, a frequency band that is used, a time slot that is used, a modulation/demodulation scheme that is used, an encryption/decryption scheme that is used, etc.), etc.

[0042] To facilitate the analyses and recommendation/suggestion generation activities described above, reference may now be made to the flowchart of a method **200d** shown in FIG. 2D. The method **200d** may be facilitated, in whole or in part, in conjunction with one or more systems, devices, or components, such as the systems, devices, and components described herein. In some embodiments, the method **200d** may be facilitated via an execution of instructions by a processing system including one or more processors. The instructions may be stored as part of one or more memories, computer or machine-readable media, etc.

[0043] In block **202d**, test data may be obtained. For example, as part of block **202d** the test data may be generated based on experience or knowledge of a universe of patterns or behaviors in respect of voice communications.

[0044] In block **206d**, one or more CAUs may be extracted from the data of block **202d**. For example, as part of block **206d**, the test data of block **202d** may be segmented into discrete units in accordance with a time interval.

[0045] In block **210d**, the data that is available via the CAUs (of block **206d**) may be labeled. For example, the labeling may serve to identify the data as speech, silence, etc. The labeling of block **210d** may be based on one or more characteristics of the data.

[0046] In block **214d**, a model may be trained. Training in this context may include a generation of the model in the first instance and/or an adaptation or modification made against/to an existing model in other instances.

[0047] In block **218d**, the model that is trained as part of block **218d** may be saved or stored. In some embodiments, one or more timestamps or versions may be assigned to the model to facilitate record keeping, audits, and the like.

[0048] In block **222d**, actual data may be obtained. For example, the actual data may be associated with a voice call or voice communication session, potentially involving one or more users or subscribers.

[0049] In block **226d**, a rule check may be performed against the actual data of block **222d**. For example, the rule check may be used to discard erroneous data, filter spurious data or anomalies, etc.

[0050] In block **230d**, a prediction may be generated regarding whether the actual data of block **222d** is subject to packet degradation (e.g., packet delay or packet loss). The prediction may be facilitated via a comparison of the actual data of block **222d** (as potentially subjected to the rule

check of block **226d**) against the model of block **218d**. In some embodiments, the prediction of block **230d** may include an affirmative determination (e.g., a prediction with 100% accuracy, which is to say that aspects of this disclosure may identify issues leading to packet degradation with certainty). To the extent that any such packet degradation is identified as likely having occurred, the degradation may be captured as part of block **234d**.

[0051] In block **238d**, a root cause analysis (RCA) may be performed to identify a root cause or source of any (likely or probabilistic) packet/signal degradation identified as part of blocks **230d** and **234d**. The RCA analysis of block **238d** may drive an identification of an improvement or remediation as part of block **242d**. The improvement or remediation may be implemented as part of block **242d** in respect of one or more entities or devices.

[0052] The RCA analysis of block **238d** may serve to provide data that may be labeled as part of block **210d**, which in turn may drive an update or modification to the model as part of blocks **214d** and **218d**. In this manner, the model may be a “living model” that may be modified or adapted based on knowledge or experience gained over time and with use. In this manner, the model may become more accurate, such that any errors in the model may tend to converge towards zero. Future iterations or executions of the method **200d** may make use of the latest or most current model (e.g., a modified model).

[0053] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2D, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0054] As set forth above, by reconstructing or partitioning/segmenting voice packets into CAUs and monitoring their values (via one or more probes or the like), a proactive approach to identifying and addressing call degradation issues may be obtained/realized. This ultimately may result in improved call quality and a better quality of user experience, benefiting both network/system operators and end-users alike.

[0055] By proactively detecting and addressing call degradation issues, aspects of this disclosure may help maintain consistently high-quality voice calls, leading to a better user experience and increased customer satisfaction. Continuous monitoring and optimization capabilities of this disclosure may contribute to the overall efficiency and reliability of a communication network/system, ensuring optimal utilization of network/system resources and infrastructure. Aspects of this disclosure may promote customer/subscriber/user loyalty; by providing a superior voice call experience, network/system operators and service providers can retain more customers/subscribers/users and reduce churn rates, which translates into higher revenue and profitability. Aspects of this disclosure provide an ability to accurately localize call degradation issues, potentially in real-time, thereby enabling network/system operators and service providers to quickly identify and address problem areas, reducing downtime and reducing (e.g., minimizing) the impact on end-users. Aspects of this disclosure empower network/system operators and service providers to proactively manage networks/systems, with an ability to address potential issues before they escalate and cause widespread disruptions.

[0056] Aspects of this disclosure may be used to absolve a network/system operator or service provider of culpability in respect of a degradation in signaling involving one or more communication sessions. For example, in a context where multiple network/system operators or service providers are involved, it may be the case that an issue experienced by a first network/system operator or service provider may impact a quality of communication sessions involving a second network operator or service provider. Remedies may be implemented by the second network operator or service provider in an effort to address the issue.

[0057] In some embodiments, data that is subject to analysis may be sourced or obtained from one or more locations. For example, in some embodiments data that is monitored via a probe may

correspond to user data that may be conveyed via a user plane or a data plane. Other data, such as control plane data conveyed via a control plane, may provide important markers or clues as to the health or status of a communication network or system, a communication session, etc.

[0058] Aspects of this disclosure may be applied in respect of a multitude of practical applications involving, for example, mobile, fixed, and federated Wi-Fi networks worldwide, Internet Service Provider (ISP) probes, application servers, and consumer, commercial, and governmental smart mobile devices and computers. To demonstrate, network operators and service providers can utilize aspects of this disclosure to proactively monitor and optimize their networks/systems, ensuring an excellent voice call experience for their customers. Large organizations with extensive internal communication networks can implement this solution to monitor and maintain voice call quality across their infrastructure, reducing communication disruptions and improving overall productivity. In mission-critical applications, such as emergency response and public safety communications, maintaining high-quality voice calls is often essential; the various aspects of this disclosure can help ensure reliable communication between first responders, dispatchers, and other emergency personnel. ISPs offering Voice over Internet Protocol (VoIP) services can integrate aspects of this disclosure into their networks/systems to monitor and enhance (e.g., optimize) call quality, providing a competitive advantage and attracting more customers in the process.

[0059] As the foregoing demonstrates, the various aspects of this disclosure are not directed to abstract ideas. To the contrary, the various aspects of this disclosure are directed to, and integrated within, various practical applications representing an improvement to technology, such that the various aspects of this disclosure are directed to significantly more than any abstract idea standing alone.

[0060] Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of system **200a**, and method **200d** presented in FIGS. 1, 2A, and 2D. For example, the virtualized communication network **300** can facilitate, in whole or in part, obtaining a model in respect of voice communications, obtaining data representative of a voice communication session, comparing the data to the model to generate a prediction that the data is subject to degradation, based on the prediction, identifying a remediation to address the degradation, and implementing the remediation. The virtualized communication network **300** can facilitate, in whole or in part, analyzing first data from a voice communication session, determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay, and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold. The virtualized communication network **300** can facilitate, in whole or in part, analyzing, by a processing system including a processor, at least first data from a voice communication session, generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold, identifying, by the processing system and based on the generating, a source of the degradation, and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.

[0061] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements

including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0062] In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330, 332, 334**, etc. that perform some or all of the functions of network elements **150, 152, 154, 156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general-purpose processors or general-purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0063] As an example, a traditional network element **150** (shown in FIG. **1**), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it is elastic: so, the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0064] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized and might require special DSP code and analog front ends (AFEs) that do not lend themselves to implementation as VNEs **330, 332** or **334**. These network elements can be included in transport layer **350**.

[0065] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330, 332, 334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330, 332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330, 332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements do not typically need to forward large amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and which creates an elastic function with higher availability overall than its former monolithic version. These virtual network elements **330, 332, 334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0066] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330, 332, 334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular,

network workloads may have applications distributed across the virtualized network function cloud 325 and cloud computing environment 375 and in the commercial cloud or might simply orchestrate workloads supported entirely in NFV infrastructure from these third-party locations. [0067] Turning now to FIG. 4, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. 4 and the following discussion are intended to provide a brief, general description of a suitable computing environment 400 in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment 400 can be used in the implementation of network elements 150, 152, 154, 156, access terminal 112, base station or access point 122, switching device 132, media terminal 142, and/or VNEs 330, 332, 334, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, the computing environment 400 can facilitate, in whole or in part, obtaining a model in respect of voice communications, obtaining data representative of a voice communication session, comparing the data to the model to generate a prediction that the data is subject to degradation, based on the prediction, identifying a remediation to address the degradation, and implementing the remediation. The computing environment 400 can facilitate, in whole or in part, analyzing first data from a voice communication session, determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay, and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold. The computing environment 400 can facilitate, in whole or in part, analyzing, by a processing system including a processor, at least first data from a voice communication session, generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold, identifying, by the processing system and based on the generating, a source of the degradation, and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.

[0068] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0069] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0070] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0071] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media,

removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0072] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0073] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0074] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0075] With reference again to FIG. 4, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

[0076] The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

[0077] The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high-capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0078] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **402**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0079] A number of program modules can be stored in the drives and RAM **412**, comprising an operating system **430**, one or more application programs **432**, other program modules **434** and program data **436**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **412**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0080] A user can enter commands and information into the computer **402** through one or more wired/wireless input devices, e.g., a keyboard **438** and a pointing device, such as a mouse **440**. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit **404** through an input device interface **442** that can be coupled to the system bus **408**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0081] A monitor **444** or other type of display device can be also connected to the system bus **408** via an interface, such as a video adapter **446**. It will also be appreciated that in alternative embodiments, a monitor **444** can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer **402** via any communication means, including via the Internet and cloud-based networks. In addition to the monitor **444**, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0082] The computer **402** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **448**. The remote computer(s) **448** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer **402**, although, for purposes of brevity, only a remote memory/storage device **450** is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) **452** and/or larger networks, e.g., a wide area network (WAN) **454**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0083] When used in a LAN networking environment, the computer **402** can be connected to the LAN **452** through a wired and/or wireless communication network interface or adapter **456**. The adapter **456** can facilitate wired or wireless communication to the LAN **452**, which can also comprise a wireless AP disposed thereon for communicating with the adapter **456**.

[0084] When used in a WAN networking environment, the computer **402** can comprise a modem **458** or can be connected to a communications server on the WAN **454** or has other means for establishing communications over the WAN **454**, such as by way of the Internet. The modem **458**, which can be internal or external and a wired or wireless device, can be connected to the system bus **408** via the input device interface **442**. In a networked environment, program modules depicted relative to the computer **402** or portions thereof, can be stored in the remote memory/storage device **450**. It will be appreciated that the network connections shown are example and other means of

establishing a communications link between the computers can be used.

[0085] The computer **402** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0086] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0087] Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, the platform **510** can facilitate, in whole or in part, obtaining a model in respect of voice communications, obtaining data representative of a voice communication session, comparing the data to the model to generate a prediction that the data is subject to degradation, based on the prediction, identifying a remediation to address the degradation, and implementing the remediation. The platform **510** can facilitate, in whole or in part, analyzing first data from a voice communication session, determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay, and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold. The platform **510** can facilitate, in whole or in part, analyzing, by a processing system including a processor, at least first data from a voice communication session, generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold, identifying, by the processing system and based on the generating, a source of the degradation, and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.

[0088] In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited

location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

[0089] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0090] In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0091] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. **1(s)** that enhance wireless service coverage by providing more network coverage.

[0092] It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more

processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0093] In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0094] In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0095] Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, the computing device **600** can facilitate, in whole or in part, obtaining a model in respect of voice communications, obtaining data representative of a voice communication session, comparing the data to the model to generate a prediction that the data is subject to degradation, based on the prediction, identifying a remediation to address the degradation, and implementing the remediation. The computing device **600** can facilitate, in whole or in part, analyzing first data from a voice communication session, determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay, and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold. The computing device **600** can facilitate, in whole or in part, analyzing, by a processing system including a processor, at least first data from a voice communication session, generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold, identifying, by the processing system and based on the generating, a source of the degradation, and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.

[0096] The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, Wi-Fi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched

wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

[0097] The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0098] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0099] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

[0100] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0101] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

[0102] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, Wi-Fi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions,

controlling, and processing data supplied by the aforementioned components of the communication device **600**.

[0103] Other components not shown in FIG. **6** can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

[0104] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0105] In the subject specification, terms such as “store,” “storage,” “data store,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, nonvolatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SLDRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0106] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0107] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0108] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after

addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $x=(x_{\text{sub.1}}, x_{\text{sub.2}}, x_{\text{sub.3}}, x_{\text{sub.4}} \dots x_{\text{sub.n}})$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0109] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

[0110] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0111] Further, the various embodiments can be implemented as a method, apparatus or article of

manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0112] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0113] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0114] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0115] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0116] As used herein, terms such as “data storage,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media,

described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0117] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible.

Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0118] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0119] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0120] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

Claims

1. A device, comprising: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: obtaining a model in respect of voice communications;

obtaining data representative of a voice communication session; comparing the data to the model to generate a prediction that the data is subject to degradation; based on the prediction, identifying a remediation to address the degradation; and implementing the remediation.

2. The device of claim 1, wherein the operations further comprise: performing a root cause analysis to identify a source of the degradation resulting in an identified source.

3. The device of claim 2, wherein the identifying of the remediation is based on the identified source.

4. The device of claim 2, wherein the operations further comprise: modifying the model based on the performing of the root cause analysis resulting in a modified model that is different from the model.

5. The device of claim 4, wherein the operations further comprise: obtaining second data representative of a second voice communication session; comparing the second data to the modified model to generate a second prediction that the second data is subject to degradation; based on the second prediction, identifying a second remediation to address the degradation in the second data; and implementing the second remediation.

6. The device of claim 1, wherein the remediation includes an adjustment to a configuration of a resource of a network or a system that supports the voice communication session.

7. The device of claim 1, wherein the voice communication session involves a user equipment, and wherein the implementing of the remediation comprises applying the remediation to the user equipment, a base station, a server, or any combination thereof.

8. The device of claim 1, wherein the prediction that the data is subject to degradation comprises a first prediction that first data of the data is subject to a first degradation included in the degradation, and wherein the first data is associated with an uplink direction.

9. The device of claim 8, wherein the prediction that the data is subject to degradation comprises a second prediction that second data of the data is subject to a second degradation included in the degradation, and wherein the second data is associated with a downlink direction.

10. The device of claim 1, wherein the data is arranged as a plurality of packets, and wherein the degradation includes a loss of a packet included in the plurality of packets.

11. The device of claim 1, wherein the data is arranged as a plurality of packets, and wherein the degradation includes a delay in a reception of a packet included in the plurality of packets relative to a transmission of the packet in an amount that exceeds a threshold.

12. The device of claim 1, wherein the model incorporates patterns in terms of silence between speech.

13. The device of claim 1, wherein the model incorporates an expected number of packets per unit time.

14. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising: analyzing first data from a voice communication session; determining, based on the analyzing, that the first data deviates from a model in a first amount that is greater than a threshold in terms of at least one of packet loss or packet delay; and based on the determining, implementing a modification such that second data of the voice communication session deviates from the model in a second amount that is less than the threshold.

15. The non-transitory machine-readable medium of claim 14, wherein the voice communication session involves a first user equipment, and wherein the implementing of the modification comprises handing over the voice communication session from a first base station to a second base station.

16. The non-transitory machine-readable medium of claim 15, wherein the voice communication session involves a second user equipment, and wherein the deviation from the model in the first amount is based on a characteristic of a communication link between the second user equipment and a third base station.

- 17.** The non-transitory machine-readable medium of claim 16, wherein the first base station and the second base station are associated with a first network operator and the third base station is associated with a second network operator that is different from the first network operator.
- 18.** A method, comprising: analyzing, by a processing system including a processor, at least first data from a voice communication session; generating, by the processing system and based on the analyzing, a prediction that second data associated with the voice communication session is likely to experience degradation in an amount greater than a threshold; identifying, by the processing system and based on the generating, a source of the degradation; and implementing, by the processing system and based on the identifying, a modification to at least one of a user equipment or a network resource.
- 19.** The method of claim 18, further comprising: obtaining, by the processing system, the first data from a probe.
- 20.** The method of claim 18, wherein the at least first data includes second data, wherein the first data is associated with a data plane, and wherein the second data is associated with a control plane.
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